MULT20015 Elements of Quantum Computing

Practice Class 10

Welcome to Practice Class 10 of MULT20015 Elements of Quantum Computing, covering exercises relevant to the material presented on optimisation using quantum computers. This lab focusses on solving portfolio optimization problems via translation to QUBO (quadratic unconstrained binary optimisation) and then using the Quantum Approximate Optimisation Algorithm (QAOA).

This week's exercises goal is to:

- Get hands-on experience with solving finance problems using quantum algorithms
- Understand how the portfolio optimization problem is solved via QAOA
- Implement and investigate the QAOA algorithm and use the output from the algorithm to calculate the solution of the original portfolio optimization problem.

The exercises in these notes are to assist your understanding of the subject and may require time outside of the practice class to complete. This practical class assumes that you have completed practical class 9.

1 Solving a toy problem

In this question we will work to solve a problem for which the quantum cost function (which we would like to minimize) has been determined to be:

$$H = Z_1 + 2Z_1Z_2 + Z_2Z_3$$

Exercise 1.1

For each of the eight possible basis states (why eight?), fill in the following table, evaluating the cost function for each basis state:

State, i	Z_1	Z_2	Z_3	H_i
$\|000\rangle$	+1	+1	+1	4
$\|001\rangle$	+1	+1	-1	2
$\|010\rangle$				
$\|011\rangle$				
$\|100\rangle$				
$\|101\rangle$				
$\|110\rangle$				
$\|111\rangle$				

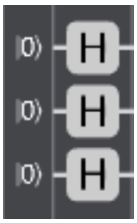
Exercise 1.2

Which state has the minimum cost?

Exercise 1.3

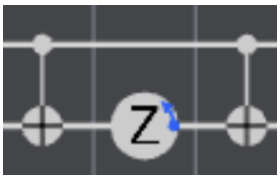
The QAOA algorithm works by constructing a "trial state" and attempts to minimize the average energy/cost function H, of the resulting state.

The first step to constructing this cost function is to put each of the qubits in equal superposition. Implement this in QUI, using Hadamard gates:



Exercise 1.4

Each of the two qubit terms has a corresponding gate sequence, implemented using QUI. Each of the two qubit gates is implemented by the combination:



The z- rotation angle used here depends on both a classical parameter, α, and the weight J_{ij} of the term in the cost function. Specifically,

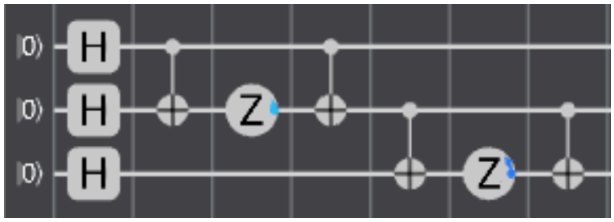
$$\theta = -\frac{J_{ij}\alpha}{2}$$

So, in our case:

$$\theta_{12} = -\frac{2\alpha}{2} = -\alpha$$

$$\theta_{23} = -\frac{\alpha}{2}$$

For $\alpha = 0.2\pi$, determine the corresponding rotation angles, θ, and implement these two sets of gates, using QUI. Your resulting circuit should look like this:



Exercise 1.5

The single qubit rotation terms in the cost function (Z variables on their own) also corresponds to a gate in the circuit to implement, as part of generating the trial state. A term in the Hamiltonian of $B_i Z$ corresponds to a single qubit z-rotation:

with an angle

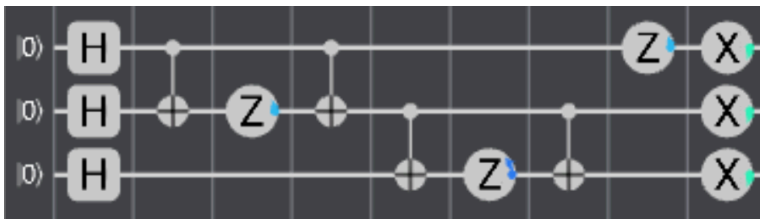
$$\theta = -\frac{\alpha B_i}{2}$$

For our cost function, evaluate the required angle for $\alpha = 0.2\pi$ and the weight of the term is $B_1 = 1$. Apply the corresponding z-rotation to your circuit.

Exercise 1.6

Finally, to complete the trial state construction, add the x block comprising of an x-rotation applied to each qubit with angle $\beta = -0.1\pi$:

The whole circuit (using just one round, k=1) should look like this:



Exercise 1.7

Use QUI to obtain a probability distribution which would be obtained by measuring the resulting state. What is the highest probability state? Does it correspond to the solution of the problem?

Exercise 1.8

Fill the following table, making use of Exercise 1.1:

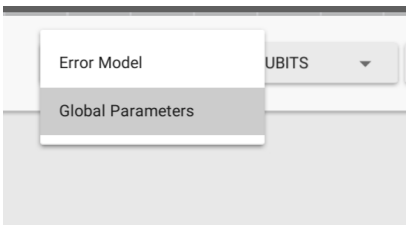
State, i	H_i	Probability, p_i	$p_i H_i$
$\|000\rangle$	4		
$\|001\rangle$	2		
$\|010\rangle$			
$\|011\rangle$			
$\|100\rangle$			
$\|101\rangle$			
$\|110\rangle$			
$\|111\rangle$			

What is the resulting expectation (ie. average value) of the cost function? To find this, sum the final column.

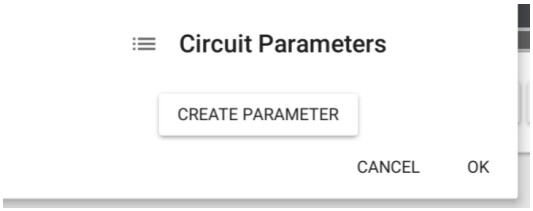
Exercise 1.9

We will now adjust the values of α and β, to minimize the expected value of the cost function and attempt to produce the state which has the highest probability of being in the solution state.

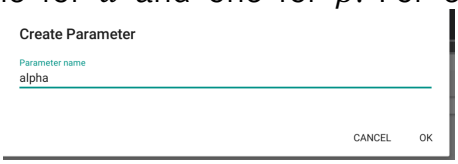
To do this, it is useful to use *global parameters* in QUI. These can be found under the Settings menu:



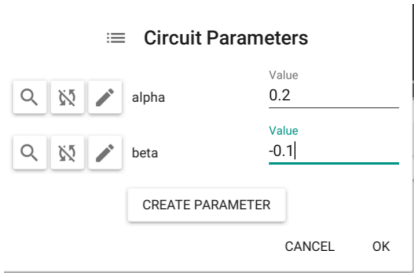
Select Global Parameters, and then “Create Parameter”



Add two global parameters, one for α and one for β. For example:

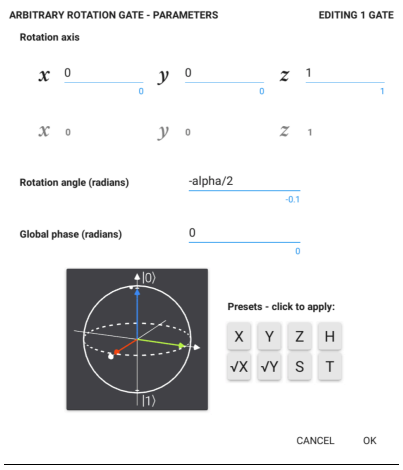


and set the corresponding values:



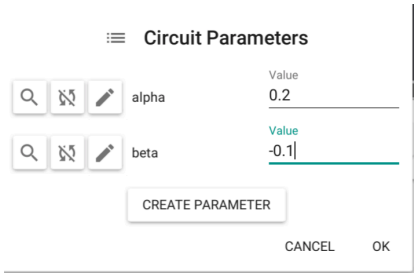
Exercise 1.10

Now edit each of your rotation gates making use of the global parameters.



Exercise 1.11

Making use of the global parameters dialog box, adjust the values, attempting to find the optimum values, leading to a smaller average value of the cost function, and the highest probability of measuring the solution state.



What are the best values of α and β which you are able to find? What is the probability of measuring the solution state for these values?

2 Portfolio optimization

The portfolio optimization problem involves computing the weights w_i such that the value of the portfolio is maximised. The value of the portfolio is given by the expression below, where:

$$\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \sigma_{ij}$$
$$\text{subject to } \sum_{i=1}^{n} w_i = 1$$

- r_i is the expected return of asset i
- λ is the penalty for risk, and
- σ_{ij} is the covariance matrix

Exercise 2.1 Consider a two asset problem where the expected returns of assets A and B are 1% and 6%, respectively, their covariance matrix is as shown below. The risk penalty is 20.

	Asset A	Asset B
Asset A	0.4	-0.2
Asset B	-0.2	0.6

Complete the table below by calculating the return, risk, and the value of the portfolio for the given weights in the first column.

Weights	Portfolio return $\sum_{i=1}^{n} w_i r_i$	Portfolio risk $\sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij}$	Portfolio value $\sum_{i=1}^{n} w_i r_i - \lambda \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij}$
$w_A = 1, w_B = 0$			
$w_A = 0.5, w_B = 0.5$			
$w_A = 0, w_B = 1$			

Exercise 2.2 Which weight combination has the highest portfolio value?

3 Portfolio optimization via QAOA

To solve the portfolio optimization problem via QAOA, we first translated the initial problem to QUBO. This resulted in the following expression, where

$$\min \quad \lambda \sum_{i=1}^{m}\sum_{j=1}^{m} x_i x_j \sigma_{ij} - \sum_{i=1}^{m} x_i r_i + \omega \left(\sum_{i=1}^{m} x_i - B \right)^2$$

- $r_i, \lambda, \sigma_{ij}$ are the asset returns, risk penalty, and the covariances, resp.
- B is the total budget, and
- ω is the penalty for violating the budget constraint.

We will continue working with the two asset example from the previous exercise. Let the budget to be \$2 and the penalty for violating the budget constraint be 5.

Exercise 3.1 Calculate the number of qubits required for the QAOA circuit.
Hint: You should do this by looking at the number of assets and the budget.

Exercise 3.2 How many qubits will be required if the number of assets was 3 and the budget was \$4?

For our translation for the two asset example to QUBO, we get the expression E_X:

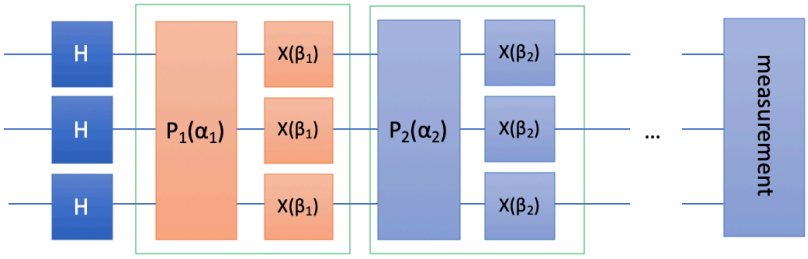
$$26x_1 1x_2 + 2x_1 x_3 + 2x_1 x_4 + 2x2x3 + 2x2x4 + 34x3x4 - 8x1 - 8x2 - 9x3 - 9x4 + 20.0$$

Exercise 3.3 Translate the $26x_3$ term to its corresponding z expression by making the substitution $x_i = (z_i + 1)/2$.

Substituting x_i with z_i in expression E_X yields the following expression E_Z (ignoring the constant term):

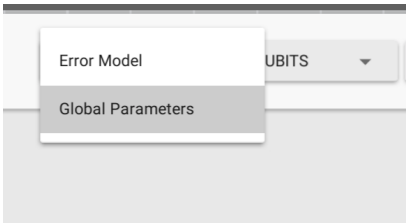
$$6.5z1z2 + 0.5z1z3 + 0.5z1z4 + 0.5z2z3 + 0.5z2z4 + 8.5z3z4 + 3.5z1 + 3.5z2 + 5.0z3 + 5.0z4$$

We will now build a QAOA circuit to minimize this expression. Recall that a QAOA algorithm has the following structure:

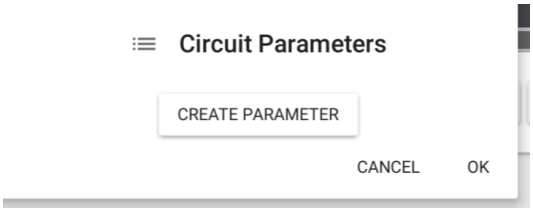


Where the phase blocks $P(\alpha)$ contain "ZZ" and "Z" snippet circuits corresponding to quadratic and unitary terms in the quantum cost function. The X block applies a x rotation to all qubits. Since the angles α and β are an input to the algorithm, it is better to add them as global parameters in QUI.

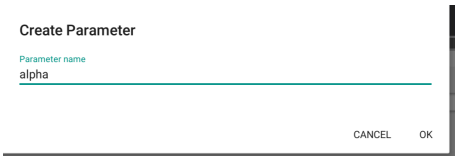
To do this, it is useful to use *global parameters* in QUI. These can be found under the Settings menu:



Select Global Parameters, and then "Create Parameter"



Add two global parameters, one for α (call it alpha) and one for β (call it beta). For example:



Set values -0.3 and -0.36 for parameters alpha and beta, respectively.

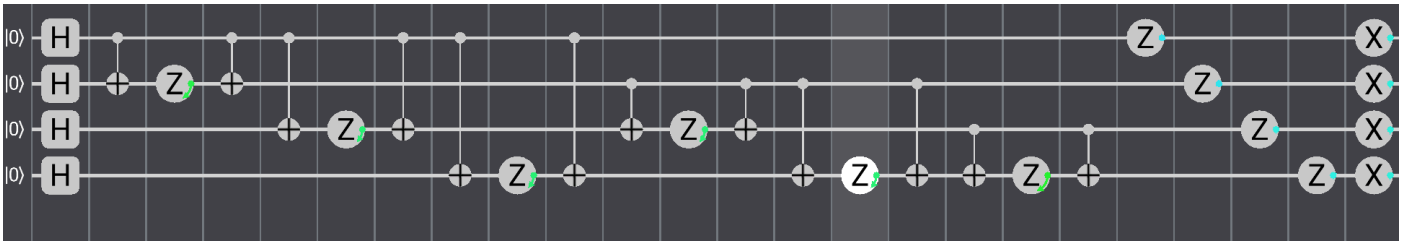
Exercise 3.4 Construct a QAOA circuit to minimize the E_Z expression.
Note that for a quadratic term $J_{12}Z_1Z_2$ where J_{12} is defined by the original optimisation problem. The corresponding angle of rotation for this term is:

$$\theta = -\frac{J_{12}\alpha}{2}$$

and for a unitary term $B_i Z$ the corresponding angle of rotation is:

$$\theta = -\frac{\alpha B_i}{2}$$

Implement the circuit for k=1 iteration. The structure should look like below.



Exercise 3.5 Compute the cost function for the state $|0110\rangle$ by substituting the values of z_i's in the expression E_Z.

Exercise 3.6 Try different combinations of angles for α and β (some suggestions are in the first column of the table below). For each of these combinations select a state with highest probability and then calculate the energy for that state.

α, β	Highest probability state	Value of cost function
$\alpha = 1.2, \beta = 0.5$		
$\alpha = 1.8, \beta = 1$		
$\alpha = -1.8, \beta = 0.5$		

4 From QAOA to asset weights in the optimal portfolio.

Our original problem required calculating the asset weight that maximise the portfolio value. We will now calculate these weights using the output from the QAOA algorithm. Recall that from the state vector one can compute the value of $x_i's$ by flipping the corresponding state. For example state $|0110\rangle$ corresponds to $x_1 = 1, x_2 = 0, x_3 = 0, x_4 = 1$. Given that the budget is \$2, x_1, x_2 are the dollars invested in asset A and x_3, x_4 are dollars invested in asset B.

Exercise 4.1 Complete the following table by calculating the asset weights from the output of QAOA obtained by running the algorithm with values of α and β as in the previous exercise.

α, β	Highest prob state	Asset weights	Portfolio value
$\alpha = 0.1, \beta = 0.1$			
$\alpha = 0.2, \beta = 0.3$			

$\alpha = -0.8, \beta = -0.3$			

Exercise 4.2 Do you see any relation between the cost function and the portfolio value?

Exercise 4.3 From whatever your best choice is of α and β, add another QAOA iteration (using a parameterised circuit with global parameters) and see if you can obtain a better solution.

Then calculate the value of the portfolio by first calculating the amount invested in each asset, and the computing the asset weights.